

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
Spring Mid-Semester Examination 2023

Date of Examination: 15.2.2023 Time: 2:00 PM - 4:00 PM Duration: 2 hours Full Marks: 30

Subject: Advanced Heat Transfer

Subject No.: CH61014

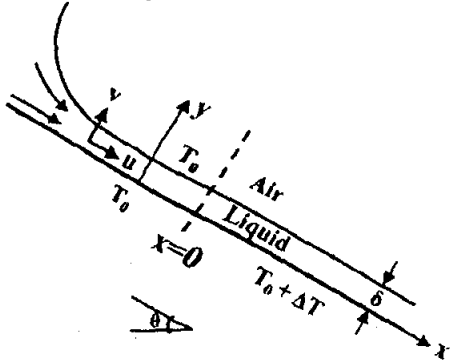
Department: Chemical Engineering.

Specific Charts, Graph paper, etc. not required

Instructions: 1. Attempt ALL questions. 2. Assume, if necessary, clearly stating them. **No queries will be entertained during the examination.** 3. Write the answers to each question in one place. 4) Credit will be given for stepwise calculations and neatness.

Part A

1. Consider the laminar flow of a two-dimensional water film on a flat wall inclined at an angle θ relative to the horizontal direction. The film flow is driven by the gravitational acceleration component acting parallel to the wall. Attach the Cartesian system of coordinates (x, y) and (u, v) to the wall, such that x and u point in the flow direction (as presented in the Figure). In this notation,



(i) Derive the *terminal velocity* distribution in the liquid film $u(y)$; in other words, determine the flow in the limit where the film inertia is negligible and the x momentum equation expresses a balance between film weight and wall friction. Let U_0 be the undetermined free surface velocity at $y = \delta$, where δ is the film thickness that remains constant. Note that U_0 is undetermined because the film flow rate can be varied.

Consider next the heat transfer from the wall to the liquid film in the case where the film and wall temperature is T_0 everywhere upstream of $x = 0$. The wall temperature alone is raised to $(T_0 + \Delta T)$ downstream of $x = 0$. Let δ_T

be the thermal boundary layer thickness of the thin liquid region in which the wall heating effect is felt.

(ii) Using scale analysis, find the order of boundary layer thickness δ_T immediately downstream from $x=0$ (where $\delta_T \ll \delta$) as a function of δ and x . Viscous dissipation effects can be neglected.

(iii) Determine the variation of $R (\delta_T / \delta)$ in the liquid film based on an integral analysis, assuming the following temperature profile:

$$\frac{T(x, y) - T_0}{\Delta T} = 1 - \frac{y}{\delta_T(x)}, \quad 0 \leq y \leq \delta_T(x) \quad \text{and} \quad T(x, y) = T_0, \quad \delta_T(x) \leq y \leq \delta$$

(iv) Up to what distance from the point $x=0$ (upstream/downstream) the integral solution will be valid?

[2+2+4+2=10]

2. (i) For steady-state two-dimensional convection of a constant-property homogeneous incompressible fluid, with no internal heat generation and negligible viscous dissipation, show that the net flow of energy (thermal diffusion and enthalpy flow) is conserved along a heatline.

(ii) Write the heatfunction that is valid inside the laminar thermal boundary layer region. [3+2=5]

Equations:

The total energy conservation and momentum balance are given as

$$\rho c_p \frac{DT}{Dt} = \nabla \cdot (k \nabla T) + q''' + \beta T \frac{DP}{Dt} - [\nabla \cdot (\tau \cdot \mathbf{v})]; \quad \rho \frac{D\mathbf{v}}{Dt} = -\nabla P + \mu \nabla^2 \mathbf{v} + \mathbf{F}$$

Where $\mathbf{F}$ is the body force vector per unit volume.

Integral boundary layer equation for momentum and energy

$$\int_0^y u(U_\infty - u) dy = \frac{1}{\rho} \gamma \frac{dP_\infty}{dx} + \frac{dU_\infty}{dx} \int_0^y u dy + v \left(\frac{\partial u}{\partial y} \right)_{y=0}$$

$$\int_0^y u(T_\infty - T) dy = \frac{dT_\infty}{dx} \int_0^y u dy + \alpha \left(\frac{\partial T}{\partial y} \right)_{y=0}$$

All the symbols stand for conventional variables and constants.

Please Turn Over

PART - B

1. Consider the steady-state heat conduction in a slab of thickness L with uniform volumetric heat source. The surface at $x = 0$ is insulated and the surface at $x = L$ is fixed at T_w . The thermal conductivity of the material (k) is a function of temperature (T) as given by $k(T) = k_0(1 + \beta T)$ where k_0, β are constants.
 - a) Write the nonlinear steady-state heat conduction equation and boundary conditions for the system. [2]
 - b) Use Kirchhoff transformation to convert this nonlinear problem (both differential equation and boundary conditions) into a linear problem. [3]

2. Consider a plane wall (without any heat source and constant thermo-physical properties) of thickness L , separating two fluids having temperature T_1 and T_2 where $T_1 > T_2$ (Fig. A, symbols carry their usual significance). Plot schematically the steady-state temperature profile between Points 1 and 2 for (a) Biot number, $Bi \rightarrow 0$, (b) $Bi \rightarrow \infty$. [2]

3. Prove that the solution of the following heat conduction problem (stationary solid with uniform heat source, specified initial condition, adiabatic boundary surfaces) is unique. (Symbols carry their usual significance. $\mathbf{r}$ is position vector). [4]

$$\frac{1}{\alpha} \frac{\partial T(\mathbf{r}, t)}{\partial t} = \nabla^2 T(\mathbf{r}, t) + \frac{\dot{q}}{k}, \quad T(\mathbf{r}, t) = T_0(\mathbf{r}), \quad \frac{\partial T(\mathbf{r}_s, t)}{\partial n} = 0$$

4. Consider the extrusion of plastics where a sheet of uniform cross-sectional area is being drawn with a constant velocity v through two rollers kept at a constant temperature T_0 (Fig. B). Heat is transferred along the length of the sheet by conduction and the sheet is cooled down by the surrounding air (heat transfer coefficient = h and temperature = T_∞) by convection. The sheet also loses heat to surrounding (temperature = T_{surf}) by radiation. Derive the steady state energy balance equation for this system. [4]

